



Afeef Kallanthodan

Embedded Software Engineer
| Motor Control | AUTOSAR

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- Indian
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About Me

Embedded SW engineer with 8+ years' experience across automotive (AUTOSAR, ISO 26262, FOC motor control) and aerospace R&D (system integration) domains. B1 certified in German.

Languages

- English
- German(B1)

Skills

- Embedded C / AUTOSAR / MISRA
- MBD - Matlab/Simulink/m-Scripting
- SW Testing - MUT/MIL/SIL/HIL
- SW Requirements (DOORS/Polarion)
- Python / Test Automation

Hobbies

- Real Simulation Games
- Read/watch latest technologies
- AI Enthusiast / Implementing local AI

Working Experience

- Sep'24–Now

Embedded SW Development/Test Engineer
Koosys GmbH | Regensburg

2 year 0 months

 - Developed MBD-based FOC algorithms for 6-phase IPMSM; generated AUTOSAR ASIL-B compliant code via Embedded Coder.
 - Built 6-phase IPMSM Plant Model and MIL environment for performance validation; tuned control parameters on MIL and real HW.
 - Created integration test environment (CANoe/WinIDEA/C#), Rest Bus Simulation, and L/R parameter estimation on real HW (SWE5/SWE6).
 - Automated unit test generation (Python), CI-CD pipeline setup, and XCP-based SW debugging.

Tools: MATLAB/Simulink, Embedded Coder, XCP, CANoe, WinIdea, CppUTest, Helix QAC, Polarion, Git, Python, C/C#
- Jul'23–Aug'24

Embedded Software/Test Engineer
Valentum Engineering GmbH | Regensburg
External Employee at Koosys GmbH | Regensburg

1 year 1 month

 - Conducted unit, integration, and system testing (SWE4–SWE6).
 - Performed static code analysis with Helix QAC and automated testing with Java/CAPL.
 - Debugged hardware (WinIDEA) and simulated networks with CANoe.

Tools: Eclipse, Java, TESSY, WinIdea, CANoe 17, Git, Helix QAC, CppUTest, DOORS
- Jul'21–Mar'23

Model Based Design Engineer
TATA Elxsi Ltd. | Bangalore

1 year 8 months

 - Developed AUTOSAR-based EPS components(AASIL-D) with MBD (Simulink/Stateflow) and unit testing.
 - Generated C code, performed static model analysis (MXAM/Polyspace), and validated via MIL/SIL.
 - Ensured compliance with ASPICE, ISO 26262, MISRA C, MAAB; Agile environment focus.

Tools: MATLAB/Simulink, StateFlow, MXAM, Polyspace, Embedded Coder, TASK-ING, GHS, Davinci, Reactis, DOORS/DNG, RTC, Synergy, Git, JIRA, Python
- Sep'17–Jul'21

Aerospace Systems Engineer
Aeronautical Development Agency (ADA) | Bangalore, India

3 years 10 months

 - Aircraft-level system integration for 5th-gen fighter program (Program Office); coordinated hydraulic, ECS and electrical subsystems across multi-disciplinary teams.
 - Evaluated subsystem architectures; resolved interface conflicts and conducted feasibility studies for system-level integration.
 - Managed bidirectional requirements traceability — aircraft-level HLRs to subsystem specifications using DOORS.
 - Developed MATLAB/Simulink/Simscape simulation models for subsystem behaviour analysis; automated mass property scripts.
 - Mechanical integration studies (weapon bay, landing gear) in Catia V5;

Tools: MATLAB/Simulink, Simscape, Stateflow, Catia V5, Teamcenter (PLM), DOORS

Standards: MIL-STD-810 **Self-Study:**DO-178C,ARP 4754A,ARP 4761A [AFuzion, DO-178CDevelopment of Safety-Critical Software - Leanna Rierson]

Education

- University Education

2011 – 2015

Bachelors in Mechanical Engineering
Focus: Mechatronics and control system, engineering mathematics, programming in C, etc.

MG University

Awards

- 2022

Extra Mile Award
Awarded by TATA Elxsi Ltd for delivering two complex and critical software components on scheduled time.

TATA Elxsi Ltd